React – Fundamentals

Session – 9

1. Install react-router-dom in your existing React app and set up BrowserRouter in your App.js file to enable routing.

Ans.

Install the package:

```
npm install react-router-dom
```

index.js:

```
import React from "react";
import ReactDOM from "react-dom/client";
import { BrowserRouter } from "react-router-dom";
import App from "./App";

const root = ReactDOM.createRoot(document.getElementById("root"));

root.render(
  <BrowserRouter>
    <App />
  </BrowserRouter>
);
```

2. Create three components: HomePage.js, DealsPage.js, and CartPage.js. Configure Routes in App.js so that visiting /, /deals, or /cart shows the correct component.

Ans.

HomePage.js:

```
function HomePage() {
  return <h2>Welcome to Home Page</h2>;
}

export default HomePage;
```

DealsPage.js:

```
function DealsPage() {
  return <h2>Today's Best Deals</h2>;
}

export default DealsPage;
```

CartPage.js:

```
function CartPage() {
  return <h2>Your Shopping Cart</h2>;
}

export default CartPage;
```

App.js:

```
import { Routes, Route } from "react-router-dom";
import HomePage from "./HomePage";
import DealsPage from "./DealsPage";
import CartPage from "./CartPage";

function App() {
  return (
    <Routes>
      <Route path="/" element={<HomePage />} />
      <Route path="/deals" element={<DealsPage />} />
      <Route path="/cart" element={<CartPage />} />
    </Routes>
  );
}

export default App;
```

3. Add a navigation bar using NavLink that lets users switch between Home, Deals, and Cart pages, and highlight the active link with a different color.

Hint: Use the 'activeClassName' or 'style' prop of NavLink to apply styles to the active link.

Ans.

App.js:

```
import { NavLink, Routes, Route } from "react-router-dom";
import HomePage from "./HomePage";
import DealsPage from "./DealsPage";
import CartPage from "./CartPage";

function App() {
  return (
    <div>
      <nav>
        <NavLink
          to="/"
          style={({ isActive }) => ({
```

```
            color: isActive ? "red" : "black",
            marginRight: "20px",
          })}
        >
          Home
        </NavLink>

        <NavLink
          to="/deals"
          style={({ isActive }) => ({
            color: isActive ? "red" : "black",
            marginRight: "20px",
          })}
        >
          Deals
        </NavLink>

        <NavLink
          to="/cart"
          style={({ isActive }) => ({
            color: isActive ? "red" : "black",
          })}
        >
          Cart
        </NavLink>
      </nav>

      <Routes>
        <Route path="/" element={<HomePage />} />
        <Route path="/deals" element={<DealsPage />} />
        <Route path="/cart" element={<CartPage />} />
      </Routes>
    </div>
  );
}

export default App;
```

4. Simulate a Flipkart-style navigation: add a NotFound.js component and configure a route so that any unknown path shows a '404 - Page Not Found' message.

Ans.

NotFound.js:

```
function NotFound() {
  return <h2>404 - Page Not Found</h2>;
}
```

```
export default NotFound;
```

Updated App.js:

```
import { NavLink, Routes, Route } from "react-router-dom";
import HomePage from "./HomePage";
import DealsPage from "./DealsPage";
import CartPage from "./CartPage";
import NotFound from "./NotFound";

function App() {
  return (
    <div>
      <nav>
        <NavLink
          to="/"
          style={({ isActive }) => ({
            color: isActive ? "red" : "black",
            marginRight: "20px",
          })}
        >
          Home
        </NavLink>

        <NavLink
          to="/deals"
          style={({ isActive }) => ({
            color: isActive ? "red" : "black",
            marginRight: "20px",
          })}
        >
          Deals
        </NavLink>

        <NavLink
          to="/cart"
          style={({ isActive }) => ({
            color: isActive ? "red" : "black",
          })}
        >
          Cart
        </NavLink>
      </nav>

      <Routes>
        <Route path="/" element={<HomePage />} />
        <Route path="/deals" element={<DealsPage />} />
        <Route path="/cart" element={<CartPage />} />
```

```
        <Route path="*" element={<NotFound />} />
      </Routes>
    </div>
  );
}

export default App;
```